Annotation: Mathematics as a Precise Instance of a General Human Practice

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Abstract

Humans annotate the world. Across the symbolic systems examined here, a recurring sequence appears: a regularity is observed, a conventional sign is fixed for it, the sign is compressed and transmitted, and eventually the sign can be mistaken for the thing it records. This paper examines that sequence through the historical development of counting and numeral notation, then compares it with spoken language and measurement. Mathematics is treated as a precise instance of this general practice rather than as a special case. The historical cases provide the primary evidence. The comparison identifies a common practice: recording observed regularities in a form that can be preserved, communicated, and extended beyond the original observation. The paper argues that the effectiveness of mathematics in describing the physical world is the expected consequence of a notation system developed by observing the thing it describes, that numeral systems are distinct representational encodings rather than competing truth-claims, and that the failure mode of the practice is representation drift, the treatment of the annotation as the thing annotated. Mathematics is not a revelation and not a fiction. It is annotation, done with exceptional care.

1 The Pattern

Across the symbolic systems examined here, a recurring sequence appears. A regularity is present and noticed. A conventional sign is fixed for it. Repeated information is compressed and encoded efficiently. The record becomes shareable and transmissible. And then, because the system is now second nature, the sign can come to feel like the thing. The word for the river starts to feel like the river. The mark for the flock starts to feel like the flock.

This paper examines that sequence through the historical development of counting and numeral notation, then compares it with spoken language and measurement. The historical cases provide the primary evidence. The comparison identifies a common practice: recording observed regularities in a form that can be preserved, communicated, and extended beyond the original observation.

The claim is not that every symbolic system has been shown to follow this sequence in full. The claim is that the sequence is a recurring pattern, visible in the cases examined here, and that mathematics is a precise instance of it. The paper is concerned with origin, with where the notation comes from and what it was built to record. It does not claim that this is the only possible account of mathematics. It claims that this is an account the historical record supports.

Under this framing, several long-standing questions have a direct answer. Why there are many numeral systems rather than one. Why mathematics, at its root, is effective in describing the physical world. Why the same quantity can be represented in different notations without changing what it is. Why old notation systems remain useful long after newer ones appear. Why notation, when it becomes second nature, is so easy to mistake for the world.

The paper proceeds in the order the pattern itself proceeds. Section 2 establishes the priority of observation over notation in the historical record of counting. Section 3 treats numeral systems as distinct representational encodings of invariant quantity. Section 4 explains how the notation extends from observation to derivation. Section 5 names representation drift as the failure mode and gives cases. Section 6 returns to what mathematics has always been: annotation of a world that was always there.

2 The Priority of Observation Over Notation

The claim that notation follows observation is not controversial once stated. It is only obscured by the habit of teaching mathematics as a finished system rather than as a historical one.

Consider what is required to count. A herder needs to know whether any animals are missing at the end of the day. The herder does not need arithmetic. The herder needs a one-to-one correspondence between a set of marks and a set of animals. The marks can be stones, notches, or lines drawn in sand. The correspondence is the operation. The notation is whatever medium is at hand.

This is an early form of recorded counting, and the use of tally marks is attested across cultures. The Ishango bone, dated to roughly 20,000 years ago, carries notched marks in grouped patterns [1]. The Lebombo bone is older still [2]. These are not mathematical treatises. This paper interprets the marks as possible records of counting, while recognizing that their precise purpose remains open to interpretation. Their significance for this argument lies in the material use of marks to preserve information, even where the information being recorded cannot now be established with certainty. They provide historical evidence of marking practices, while the progression from tally marks to shared numeral systems is supported by the broader historical record. Annotation, in other words, is what a person does to carry an observed pattern across time, or to someone who was not there to observe it, and notation is the system of signs in which the annotation is made. The pattern comes first. The annotation comes second.

The sequence runs as follows:

Observation. A recurring pattern is present in the world: the flock, the cycle, the quantity.

Tally. A one-to-one correspondence is established between the pattern and a set of marks.

Compression. When the count grows large, a single mark is assigned to a group. This is an early abstraction.

Relation. The behavior of combined, removed, or repeated groups is observed and recorded.

Numerals. Symbols are fixed so the relations can be written and reproduced.

Transmission through a shared system. The numerals allow the pattern to be communicated to anyone who learns the system.

The sequence illustrates one natural path from observation to a shared system of notation. It is not a fixed order that every culture or numeral system followed. Each stage adds a way to record, organize, or transmit what has been observed, and the next stage is the one that follows naturally from it. Notation arises from the need to record something observed. The historical examples support the progression from tally marks to numerals, and, in the systems that developed positional notation, from numerals to positional systems. The existence of ancient notches does not by itself establish what was counted or prove every stage in the proposed sequence. The qualification is part of the claim: the sequence describes a natural path, not a universal history.

The same progression is visible outside mathematics. A spoken language grows a word for a recurring feature of the environment, then compresses it into a classifier or a compound, then transmits it to speakers who were not present to observe the feature [3]. A system of measurement grows a unit for a recurring quantity, then compresses it into a prefix or a multiple, then transmits it to users who never handled the original standard. The metre shows the sequence. In 1791 the French Academy of Sciences chose to define it as one ten-millionth of the distance from the North Pole to the equator along the meridian through Paris, so that distance is 10,000,000 m = 10,000 km by definition. In 1983 it was redefined as the distance light travels in vacuum in 1/299,792,458 of a second, which fixes the speed of light at exactly 299,792,458 m/s. A kilometre compresses a multiple of the metre into a prefix, 1 km = 1000 m, and users of the unit do not

need to repeat the original measurement [4]. The mathematical case is the same sequence run with unusual precision, which is why the historical record of counting is the clearest evidence for the general claim.

Counting is correspondence by definition. The correspondence is established first, and the numerical sequence is the order along which that correspondence runs. The numerical sequence is unbounded, and its unboundedness is a property of the rule that extends the notation, not of the flock.

3 The Same Quantity, Different Notations

The existence of multiple numeral systems shows that the notation used to represent a quantity is not uniquely determined by the quantity itself. History records many numeral systems, each internally consistent, each capable of recording the same counts, and each shaped by the practical constraints of the culture that produced it.

Roman numerals write quantities with the symbols I, V, X, L, C, D, M. The numerals I, II, and III illustrate how repeated strokes can represent successive quantities. The broader Roman numeral system uses additional conventional symbols and additive or subtractive rules to represent larger numbers. The system is cumbersome for large numbers and has no native zero, but it is adequate for the administrative and architectural purposes it served.

Egyptian fractions generally expressed fractions as sums of distinct unit fractions, with special signs also used for certain fractions, including two-thirds and three-quarters. Three-quarters can be expressed as one-half plus one-quarter, because $1/2 + 1/4 = 2/4 + 1/4 = 3/4$. This illustrates how the same quantity can be represented through different notational conventions [5].

Sexagesimal, or base 60, originated in Mesopotamia and survives in the modern division of hours into minutes and minutes into seconds, and of circles into 360 degrees. The properties of 60 are one reason the system persisted. It is divisible by 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, and 60, which makes fraction handling straightforward without decimal notation [5, 6]. These are all twelve divisors of 60, since $60 = 2^2 \times 3 \times 5$ and $(2 + 1)(1 + 1)(1 + 1) = 12$.

Chinese mathematical practice used positional decimal numerals, while commercial and financial contexts also developed distinct numeral forms. Some accounting forms used more complex characters to make fraudulent alteration harder [5, 7].

Binary is the minimal positional system: two symbols, zero and one. It is the natural notation for any system with two distinguishable states, which is why it underlies digital computation. It is not more fundamental than decimal. It is more convenient for a specific class of physical implementations.

Babylonian, Mayan, and Indian numeral systems illustrate distinct developments in positional notation and the representation of zero. These developments occurred at different times and through different routes [5, 6, 7].

The pattern is consistent. Each system is a set of notational choices. How to represent quantity with symbols. How to compress repeated quantities. How to represent absence, or zero. How to represent position, or place value. How to represent operations.

Different cultures made different choices because they had different needs, different materials, and different constraints. The same quantity can be expressed in any of them. The quantity does not change when the notation changes. This is what it means for quantity to remain invariant across representational systems.

Different representational systems serve as distinct encoding conventions for recording the same underlying quantities, such that accurate translation between systems preserves the invariant quantity.

A numeral system works as a ruler works. A board that measures 30 centimeters measures $30 \div 2.54 = 3000/254 = 1500/127 \approx 11.811$ inches. The board is the same, and the two numbers are the same length written with two rulers. Mistaking the ruler for the thing it measures is the error that Section 5 names.

The number 2026 can be written in each of these systems. In decimal it is 2026, because $2 \times 1000 + 0 \times 100 + 2 \times 10 + 6 \times 1 = 2026$. In Roman numerals it is MMXXVI, because $1000 + 1000 + 10 + 10 + 5 + 1 = 2026$. In base 60 it has the two digits 33 and 46, written 33,46 with a comma separating the base-60 digits, because $33 \times 60 + 46 = 1980 + 46 = 2026$. In binary it is 11111101010, because $1024 + 512 + 256 + 128 + 64 + 32 + 8 + 2 = 2026$. The notations differ and the number is the same. The notations also differ in how efficiently their users can calculate with them.

The invariance belongs to the quantity. A finite-precision positional representation approximates quantities it cannot represent exactly. For rational quantities, the choice of base determines which have terminating expansions and which require approximation when only finitely many digits are stored. One third is written 0;20 in base 60, because the sexagesimal fractional digit 20 represents $20/60 = 1/3$. In base 10 it is 0.333…, which never terminates. One tenth is 0.1 in base 10 and 0.0001100110011… in base 2, which never terminates. The binary digits come from repeated doubling: $0.1 \times 2 = 0.2$ (digit 0), $0.2 \times 2 = 0.4$ (digit 0), $0.4 \times 2 = 0.8$ (digit 0), $0.8 \times 2 = 1.6$ (digit 1), $0.6 \times 2 = 1.2$ (digit 1), and then 0.2 returns, so the digits 0011 repeat.

The circle constant shows the same invariance in a measured quantity. The Rhind Papyrus gives the area of a circle with diameter d as the square of 8/9 of the diameter, which is $(8d/9)^2 = 64d^2/81$. Writing $d = 2r$ gives $64(2r)^2/81 = 64 \times 4r^2/81 = 256r^2/81$, and setting this equal to πr^2 gives $\pi = 256/81 = 3 + 13/81 \approx 3.1605$. Babylonian sources correspond to $25/8 = 3 + 1/8 = 3.125$, and Zu Chongzhi gave $355/113 = 3 + 16/113 \approx 3.14159$ [7]. The values differ because the methods and notations differ, and each approximates the same ratio of circumference to diameter, 3.14159265….

4 From Observation to Derivation

If mathematics is a notation system for observed pattern, then the structure of the notation should reflect the structure of what it records. It does, in a sense that can be stated precisely.

The word harmonic is used in this paper in the sense shared by its acoustic and analytic uses: structure organized by recurrence, with component frequencies occurring at integer multiples of a fundamental frequency. In acoustics, the harmonics of an ideal vibrating string have frequencies $f_n = n f_1$ for $n = 1, 2, 3, \ldots$, where $n = 1$ is the fundamental. Since a period is the reciprocal of a frequency, the periods are $T_n = 1/f_n = 1/(n f_1) = T_1/n$, where $T_1 = 1/f_1$. In analysis, a periodic phenomenon decomposes into components whose frequencies are integer multiples of a base frequency under the appropriate Fourier-series conditions. The word is not used here in the sense of the harmonic mean, harmonic functions in potential theory, or the sum of reciprocals called the harmonic series.

Recurrence alone does not establish harmonic relationships. A phenomenon can recur without its components standing in integer ratios to one another. Vibrating strings are a good example of harmonic structure because the integer ratios are established by the physics of the string. Orbital resonances can be good examples when the relevant frequency ratios are established.

Under this definition, the physical world contains harmonic structures that mathematical notation can describe. Mathematical systems can encode relationships abstracted from observed regularities, and their derived consequences can be compared with further observations. The effectiveness of mathematics in describing the physical world is therefore understandable within this framing: its representations can preserve relationships found in the phenomena they describe.

The relationship is not limited to recording patterns already observed. Once a notation system has been established, its rules permit relations to be derived that were not directly observed when the system was created. Those derived relations can then be compared with further observations. The notation can therefore extend beyond its original records without losing its connection to the world. Its symbolic conventions are human-made; the physical validity of a resulting description depends on whether the relations it expresses hold for the phenomenon being described. Higher-dimensional spaces, non-Euclidean geometries, and ab-

stract topologies are extensions of this kind: the rules of the notation permit them, and they can be constructed whether or not a physical counterpart has been found. Section 6 returns to this point. The sequence is not only observation, notation, description. It can also be observation, notation, derivation, prediction, comparison with observation.

The same reasoning explains why different numeral systems can produce equivalent results. They are distinct encodings of the same quantities, and correct translation preserves the relationships being represented. The quantity is invariant. The notation is not. When two numeral systems appear to disagree, translation can resolve the discrepancy, provided the precision of each representation is accounted for. Different mathematical frameworks can also describe different structures under different assumptions. Translation between representations must preserve the relevant conditions, not merely the words or symbols.

The harmonic case is one instance of the general point and the paper does not claim more for it than that. The world is patterned; the notation records pattern; where the pattern is recurrence with integer ratios, the notation records it with periodic functions and with the arithmetic of small integers. Where the pattern is something else, the notation records that instead. The claim is not that all of mathematics is harmonic. The claim is that the fit between notation and world is the fit between an annotation system and what it was built to annotate.

5 Representation Drift

If mathematics is annotation, then the failure mode is predictable. The failure occurs when the annotation is treated as the thing itself rather than as a record of the thing.

This failure is called representation drift. It occurs when a quantity is identified with its representation rather than with what the representation records. When a relationship is treated as a property of the notation rather than a property of the pattern. When a result obtained in one notation is assumed to hold in another without translation. When a model is treated as the system rather than as a description of the system.

Two forms of the failure are worth distinguishing. The first is encoding failure: the representation does not preserve the intended quantity correctly, as with incompatible units or finite precision. The second is scope failure: a result is applied outside the assumptions under which it holds, as when a Euclidean theorem is transferred unchanged to spherical geometry. Both are a loss of fidelity between the quantity observed and the sign that records it, and they are not identical mechanisms.

The finite-digit case in Section 3 is an instance of encoding failure. In exact arithmetic, $0.1 + 0.2 = 0.3$. In common 64-bit binary floating-point arithmetic, the sum is typically displayed as 0.30000000000000004 rather than 0.3. The discrepancy arises because the decimal inputs are approximated in binary floating-point representation, and the computation operates on those stored values.

Representation drift is not a rare error. It is a common error in any mature symbolic system, because mature systems become second nature to their users, and second nature is easy to mistake for reality. The word for the river comes to feel like the river; the mark for the flock comes to feel like the flock; the decimal expansion comes to feel like the number. The same failure pattern can appear across systems that annotate. The existence of a representation is not evidence, by itself, of the existence of its referent.

The consequences are practical. In engineering, a numerical result obtained in one unit system may be carried into another without conversion, producing an error that is attributed to the system rather than to the representation. The Mars Climate Orbiter was lost in 1999 when thruster impulse data in pound-force seconds was read by navigation software that expected newton-seconds. Since 1 lbf $\approx$ 4.448 N, each value read as newton-seconds corresponded to a physical impulse about 4.45 times smaller than the true one, and the unit mismatch propagated into navigation calculations, contributing to the loss of the spacecraft [8]. In physics as in any science, a model may be treated as the phenomenon itself, so that its assumptions, limitations, or fitted parameters are mistaken for properties of the phenomenon and not features of the model. In mathematics,

a result proven within one formal system may be treated as holding in every system, or in the world itself, without checking that the system's assumptions carry over.

The spherical triangle is an instance of scope failure. The angles of a triangle sum to 180° in Euclidean geometry. On a sphere of radius R, the triangle with three right angles has area $4\pi R^2/8 = \pi R^2/2$ and an angle excess of area/R^2 = $(\pi R^2/2)/R^2$ = $\pi/2$ radians = 90°, so its angles sum to 270° = 180° + 90°. On the Earth, taken as a sphere, this triangle has the North Pole and two points on the equator 90° apart in longitude as its vertices, and its three angles are 90° each, because the meridians meet the equator at right angles and the angle at the pole equals the 90° difference in longitude, so the 270° sum belongs to the surface and not to the notation. The word triangle is the same in both geometries and the theorem is not.

In every case, the error is the same. The map has been mistaken for the territory.

The correction at the root is not a new tool. Tools such as unit checking and dimensional analysis are formal methods for detecting certain classes of inconsistency, and they enforce translation once translation is understood to be necessary. The understanding comes first in the sense that a tool must be interpreted to be used, but people can also discover a need for translation precisely because a checking procedure reveals a mismatch. The correction at the root is remembering what the practice is. It is annotation. It records the world. It is not the world. Representational systems are encodings. They can be translated. The quantity does not change when the notation changes. When this distinction is maintained and checked, representation drift can be detected and corrected. When it is forgotten, drift can go unnoticed and propagate through subsequent work.

6 What Mathematics Has Always Been

Mathematics is annotation. The patterns it records in the physical world are observed and real. The symbols it uses are made and human. The practice connects the two, and the fidelity of that connection is what progress in describing the world measures.

This framing explains the historical record. Many numeral systems exist because many cultures made different notational choices for the same observed relationships. The same pattern appears across the symbolic systems examined here, and mathematics is a particularly precise instance because its notation permits exact specification and systematic verification of relationships. The notation can also extend beyond the observations that contributed to its formation. The ability to construct new descriptions within a notation system does not establish that every description corresponds to a physical referent. A spoken language can form a word for a unicorn without establishing that a unicorn exists. A mathematical notation can construct structures no one has measured without that construction alone establishing a physical counterpart. Origin and expressive capacity are separate things. The notation can originate in the need to record observed regularities and subsequently become capable of expressing things that were never observed.

The failure mode of the practice is representation drift. Treating the annotation as the thing itself. The correction is remembering what mathematics has always been.

It is annotation. A way to write down what we observe, so it can be carried across time and transmitted to someone who was not there. The first person who drew a line in the sand for each animal in the flock was doing mathematics. They observed a collection and recorded it with marks. Everything since has been refinement and extension. Refinement is better symbols, better systems, better instruments, better measurements, and more accurate descriptions of what we are actually observing. Extension is new structure built from the notation itself, and the notation can extend without losing its connection to the world, because derived relations can be compared with further observations. Both grow from the root this paper describes: annotation of a world that was always there.

Old systems remain useful because they were accurate records of what they recorded, within the representational purposes each was built to serve. Roman numerals still count what they always counted. Base 60

still divides time the way it always did. They were not wrong. They were distinct notational encodings. And representational choices, properly understood, are not errors. They are different systems for representing the same quantities.

Mathematics is a way to annotate the physical world, as the first person did who drew a line in the sand for each animal in the flock. The referent does not change. The description does. That is what annotation is for. Mathematics does not merely record what we already know. It also lets us derive and test things we did not know when the notation was first created.

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